12 Denise wants to give a pen set to every student in her school.
There are 799 students in the school.

Denise already has 102 pen sets.
She will need to buy more pen sets.
Each pen set costs 89 pence.

(a) Work out an estimate for the total cost of the pen sets Denise needs to buy.

Round 799 to 800 and 102 to 100.

1 mark
Sensible estimates

Estimated sets needed = 800 − 100 = 700.

Round 89p to 90p.

1 mark
Multiply estimates

Estimated cost = 700 × 90p = 63000p = £630.

1 mark
£630

(3)

(b) Is your answer to (a) an underestimate or an overestimate?
Give a reason for your answer.

Overestimate: we use 700 sets instead of 697,

1 mark
Both factors increased

and 90p instead of 89p; both make the cost higher.

(1)

(Total for Question 12 is 4 marks)